AICC TRACK



AI Computing Continuum Use Cases and Collaboration Items

Xiaojing Zhang, NTT DOCOMO Business

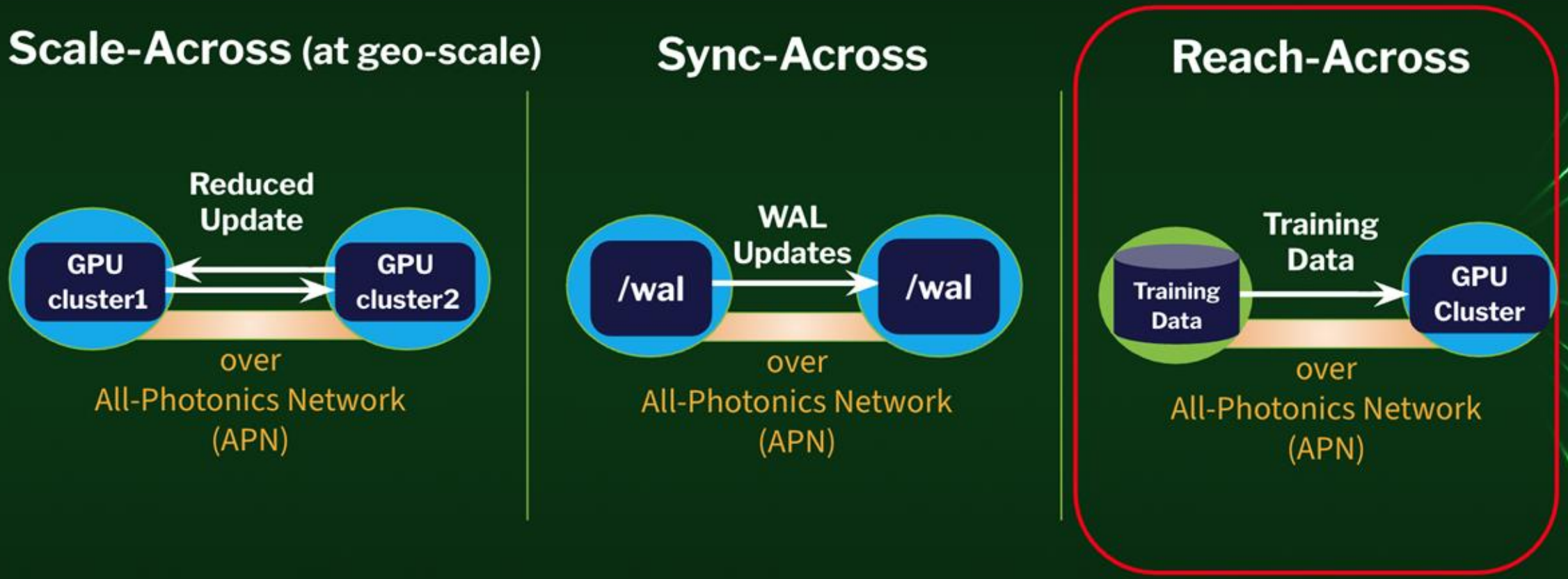
Naoto Umemori, NTT Data

Hidetsugu Sugiyama, Red Hat

Junichi Sugiyama, 1Finity

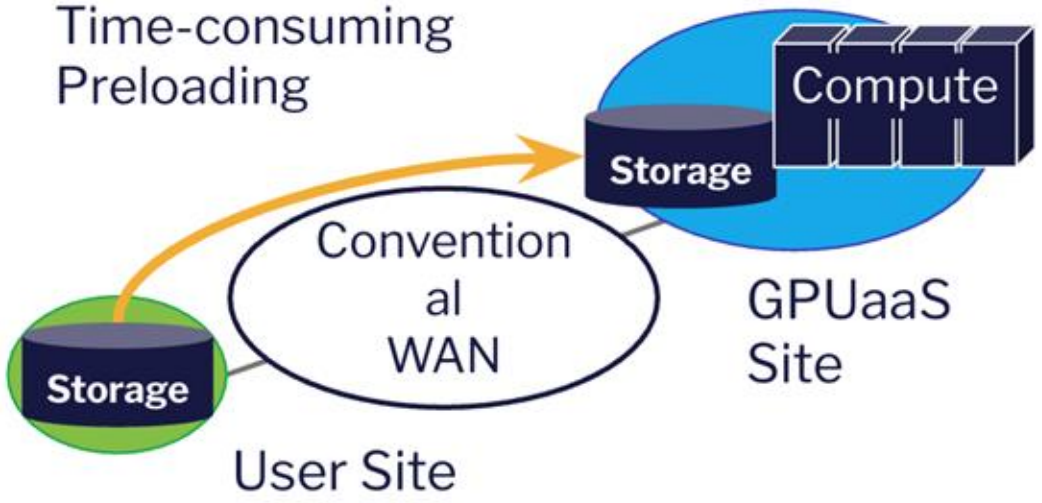
Koji Watanabe, NTT

AICC Compute-Across Models over the All-Photonics Network



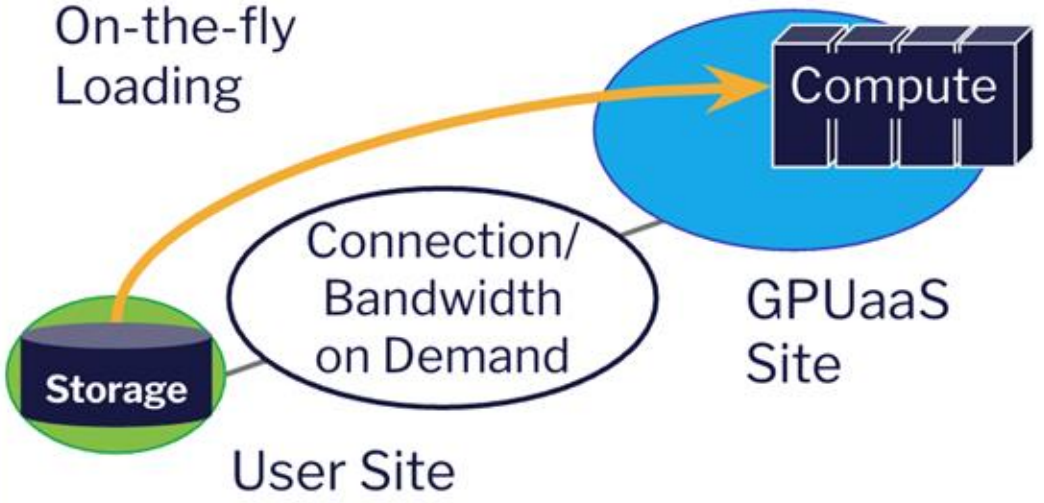
AICC Use Cases
Zero-Trust GPU Sharing for Data Sovereignty

GPU sharing today



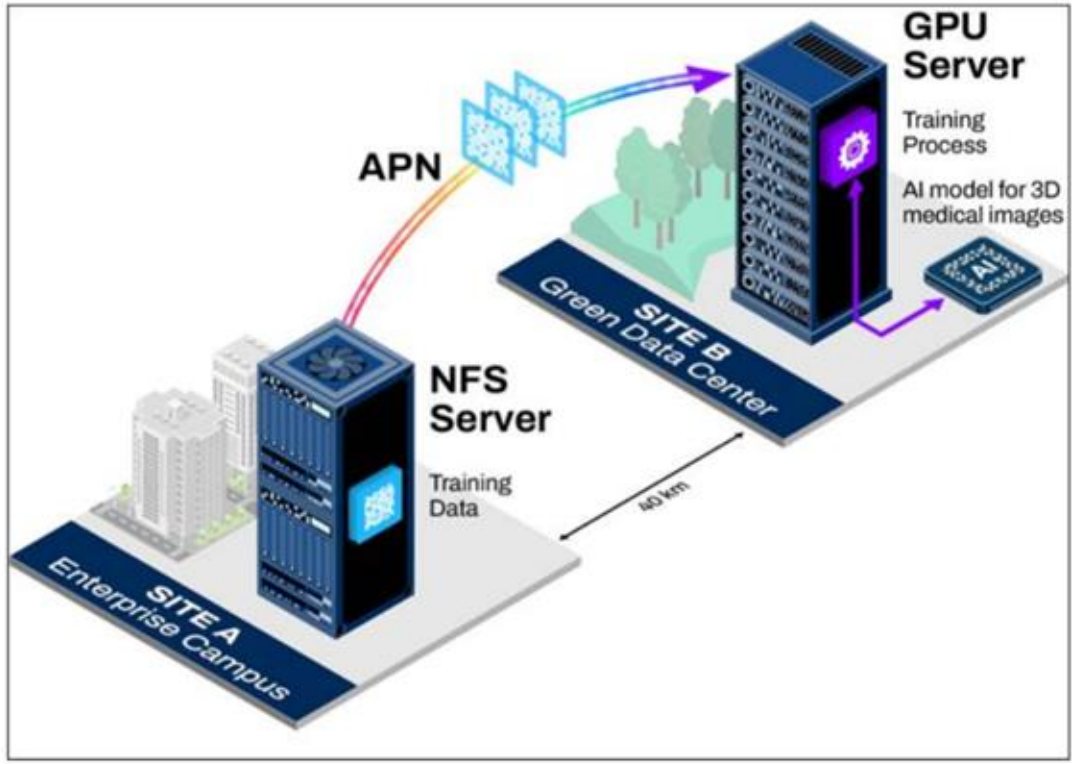
Copy data to cloud storage

GPU sharing in AICC



Keep data local

Reach-Across PoC: Remote GPU over APN — Setup



Direct Access design — keep data local (Site A)

Workload:
Unet3D (3D medical image segmentation) within MLPerf Storage v1.0.1

Dataset: 28,000 images ≈ 3.8 TB

Sites: storage at Site A (enterprise campus) GPU at Site B (green DC), NVIDIA H100 × 4

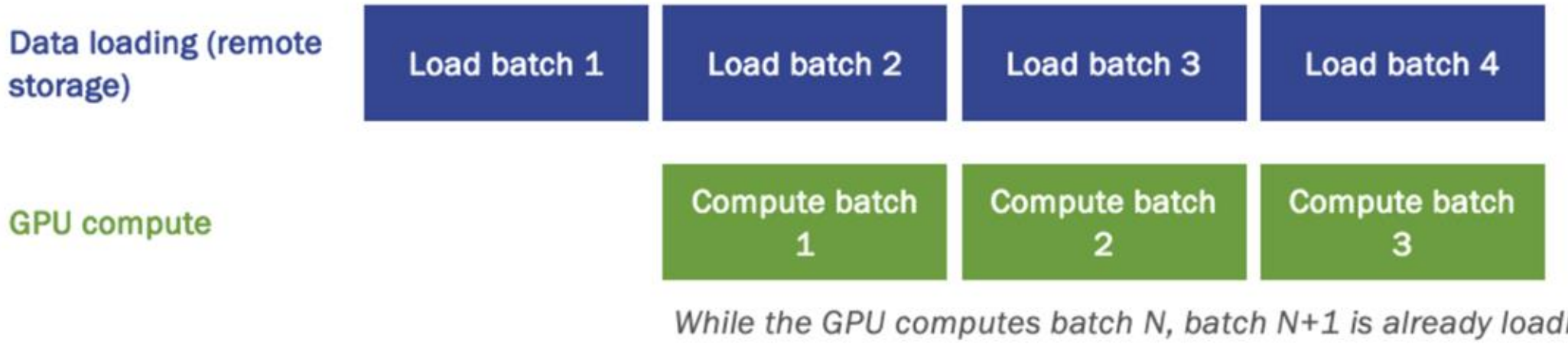
Network: 40 km physical via 100G APN; up to 500 km emulated

Storage access: NFS over TCP / RDMA

Baseline: same-site storage + GPU for comparison

Why Distance Barely Matters: Pipelined Loading

Data loading and GPU compute run **in parallel, batch by batch** — so the network hides behind computation.



While the GPU computes batch N, batch N+1 is already loading

No slowdown when: per-batch load time ≤ per-batch compute time — the WAN must **sustain throughput above the GPUs' data rate** — APN's low, stable latency is what keeps that throughput high over distance.

PoC Results → Scaling the Continuum

+1.9% **Training time**

vs local at 40 km (357 → 364 s/epoch); below +10% within 100 km

94% **Throughput kept**

at 100 km (9.66 GiB/s, NFS nconnect = 16); internet-class link ~89× slower

Parity **Total power (incl. PUE)**

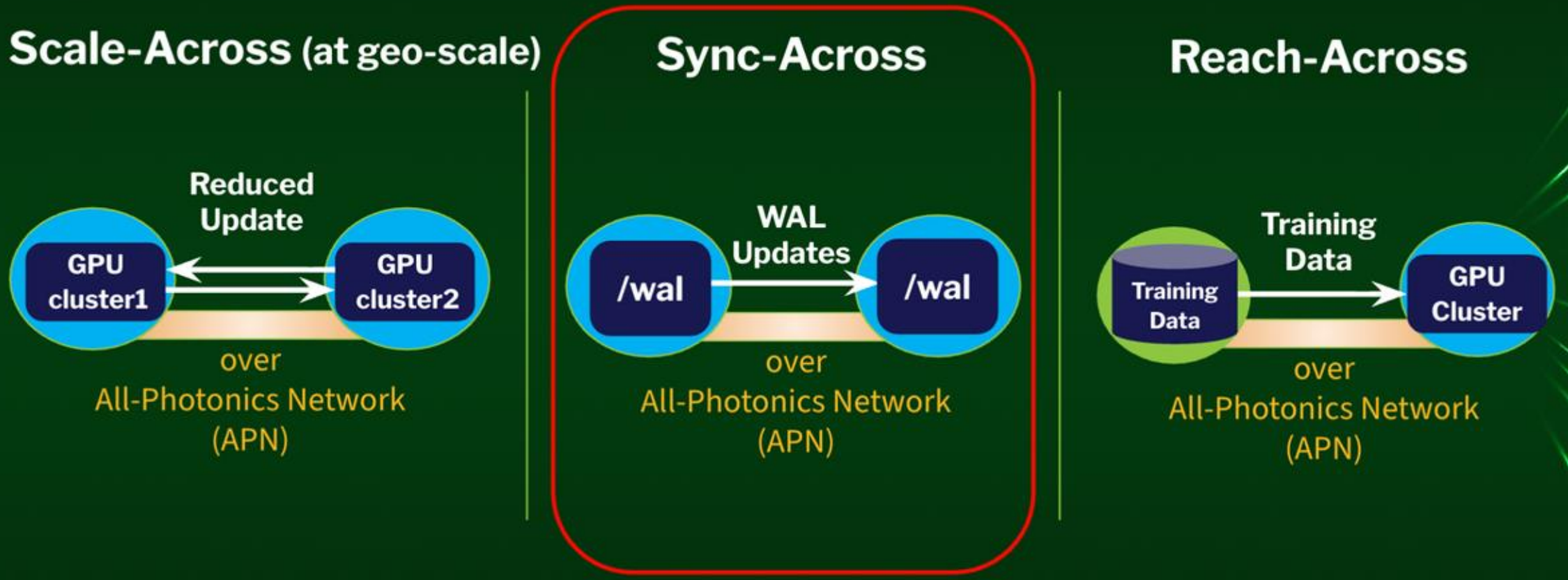
green-DC efficiency (PUE 1.56 vs 1.98) absorbs the +20% IT energy

Secure	Sustainable	Cost-efficient	Near-local
Data never leaves your site	Compute where clean energy is	NaaS instead of storage fees	Remote GPU feels like on-site

From one node to the Continuum — co-design with OCP × IOWN GF

Today: 1-node PoC → Scale-Up → Scale-Out → Scale-Across at geo-scale (APN)

AICC Compute-Across Models over the All-Photonics Network



IOWN APN-Based Financial DataCenter Interconnect

Financial Service Infrastructure - Why Change Now?

AI does not replace financial infrastructure requirements; it raises the bar for them.

What finance must never compromise



Resilience | Security | Accessibility

Financial services must remain resilient, secure, and accessible even during disruption.

- Stock services: deposits and records demand stability and integrity.
- Flow services: payments and settlements require 24/7 continuity.

AI Use in Financial Services



Automation | Personalization | Integration

- Automation: transaction monitoring, operations, and risk response.
- Personalization: customer-tailored offers and next-best actions.
- Integration: one service across banking, payments, securities, and advisory.

Future Vision of Financial Service Infrastructure

- From siloed domains to user-consented, AI-integrated service -

FROM SILOED DOMAINS

Fragmented data, limited insights, manual processes

BANKING

- Accounts
- Transactions
- Customer data

PAYMENTS

- Payments
- Settlement
- Wallets

SECURITIES

- Holdings
- Trades
- Market data

ADVISORY

- Profiles
- Recommendations
- Reports

KEY CONSIDERATIONS

Customer data is governed within each financial domain

Cross-domain insights require consent, policy, and trusted access

Siloed systems and data

Manual processes and delays

Limited customer context

Suboptimal decisions and experiences

TO USER-CONSENTED, AI-INTEGRATED SERVICE

Connected data, intelligent insights, seamless experiences

USER-CENTERED

User consent & preferences

- Transparency
- Control
- Privacy
- Security

TRUSTED ACCESS & GOVERNANCE

Proper authentication and authorization

Access only the right data of the right customer for the right purpose

Identity Verification

Consent Management

Policy Enforcement & Audit

AI SERVICE ENGINE

AI references the specific data of a specific customer appropriately

TRUST & ASSURANCE OF AI SERVICES

Internal Evaluation (System View)

- Data lineage & traceability
- Model performance & robustness
- Bias, fairness & explainability
- Continuous monitoring & alerts

External Evaluation (User View)

- Explanation for decisions
- Reliability & accuracy metrics
- User feedback & redress
- Third-party audits & certification

SEAMLESS, PERSONALIZED EXPERIENCE

One service across banking, payments, securities, and advisory

- Personalized
- Proactive
- Consistent
- Secure
- Trusted

ENABLING FOUNDATION

Secure & compliant by design

Interoperable by standard

Scalable & resilient infrastructure

Ecosystem collaboration

GREATER VALUE FOR USERS

1. Seamless cross-domain service

2. User-controlled data usage

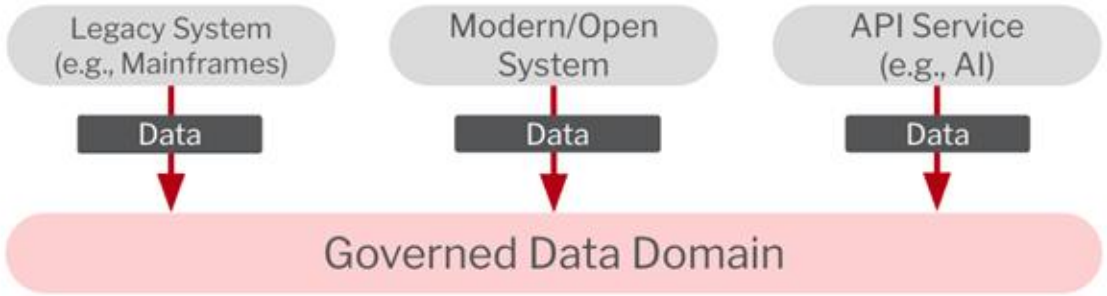
3. Near-user AI with a common committed state

IOWN GF[*1] Financial DataCenter Interconnect Use-Case Study

What IOWN APN is Expected to Enable

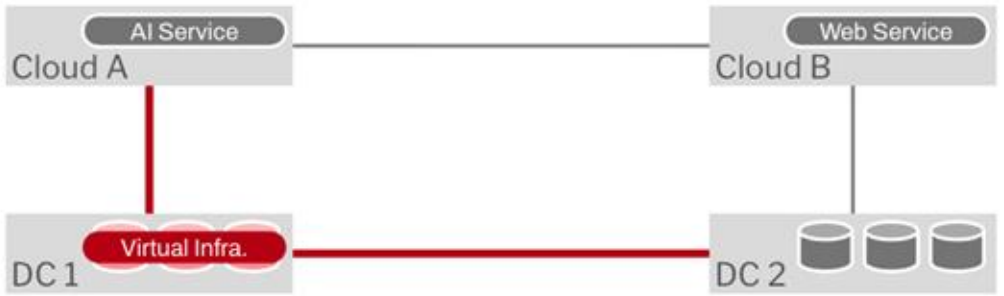
IOWN APN can enable governed data access and active use of remote resources:

Point 1: Governed Remote Data Access



IOWN APN is expected to let applications read and write remotely located data with low-latency, predictable connectivity.

Point 2: Active Use of Distant ICT resources[*2]



IOWN APN is expected to make remote compute, storage, and DC resources available as a single service fabric, not only as standby capacity.

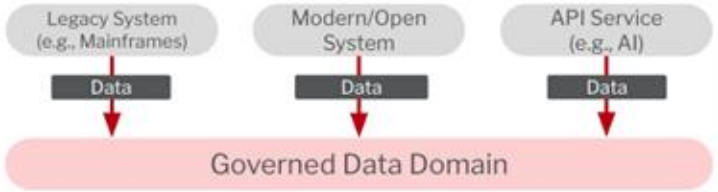
*1: IOWN Global Forum
*2: Distributed data centers and/or hybrid clouds

Pathway towards Next Generation Financial Service Infrastructure

Problem Statement

Point 1

Governed Remote Data Access



Point 2

Active Use of Distant ICT resources

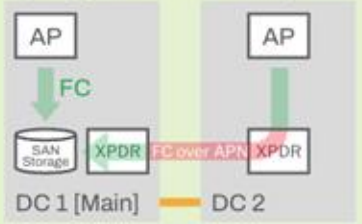


PoC – Step 1

Point 1 – PoC Step 1

FC-SAN over IOWN APN

(Long-distance FC-SAN)



Benefits

1. Secure placement flexibility
2. Better use of computing resources
3. Long-reach, lossless data access

Point 2 – PoC Step 1

Resource integration between DCs

(Long-dist. DB sync. replication)



Benefits

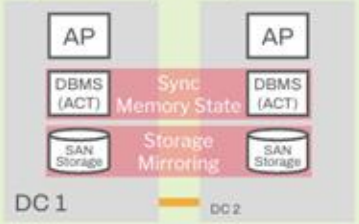
1. Precise time estimation
2. Maximizing ICR resources and facilities
3. Simplifies system architecture

PoC – Step 2

Point 1 – PoC Step 2

FC-SAN over IOWN APN

(ACT/ACT operation)



Benefits

1. Higher resource utilization
2. Faster recovery
3. Liong-distance shared state

PoC 2 – PoC Step 2

Resource integration across multiple DCs

(Distributed DB over IOWN APN)

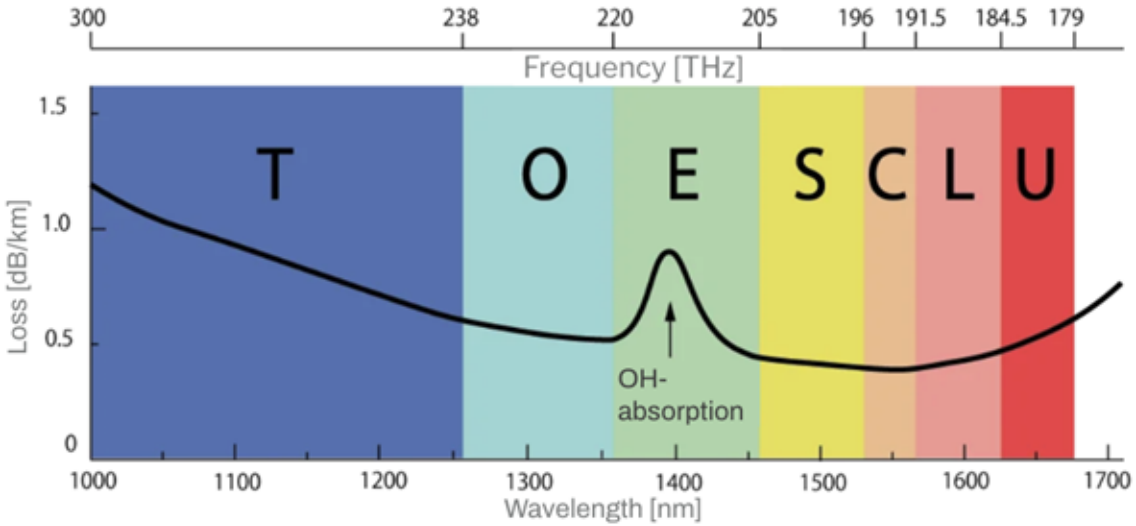


Benefits

1. Always-on service access
2. Consistent digital experience
3. Closer service, governed data

: IOWN APN

A challenge: Optical Transport Considerations for FC over APN



FC

APN

Band name		Wavelength[nm]
T-band	Thousand-band	1000-1260
O-band	Original-band	1260-1360
E-band	Extended-band	1360-1460
S-band	Short-wavelength-band	1460-1530
C-band	Conventional-band	1530-1565
L-band	Long-wavelength-band	1565-1625
U-band	Ultra Long-wavelength-band	1625-1675

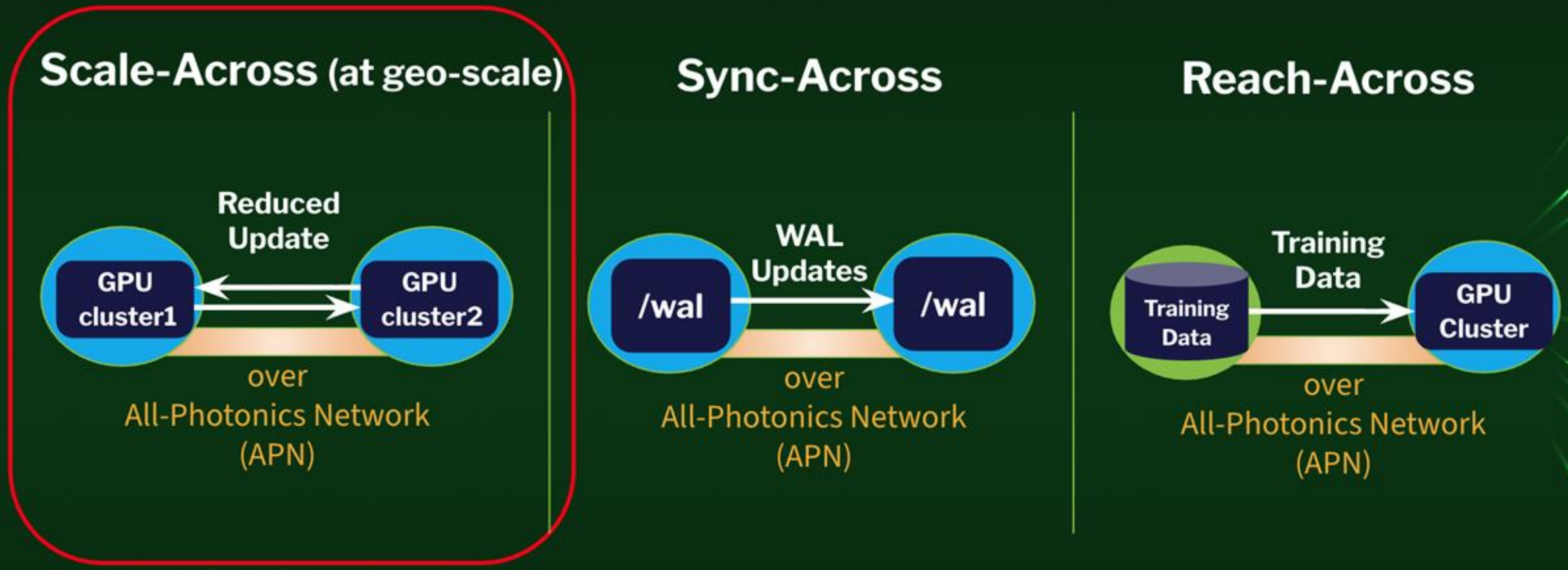
As-Is:
- FC client optics and APN/DWDM line transport use different wavelength bands. Today, these two sides are connected through a transponder.
- The transponder performs O-E-O conversion and wavelength adaption for transport.

To-Be: Ideally, the signal would remain in the optical domain without O-E-O conversion.

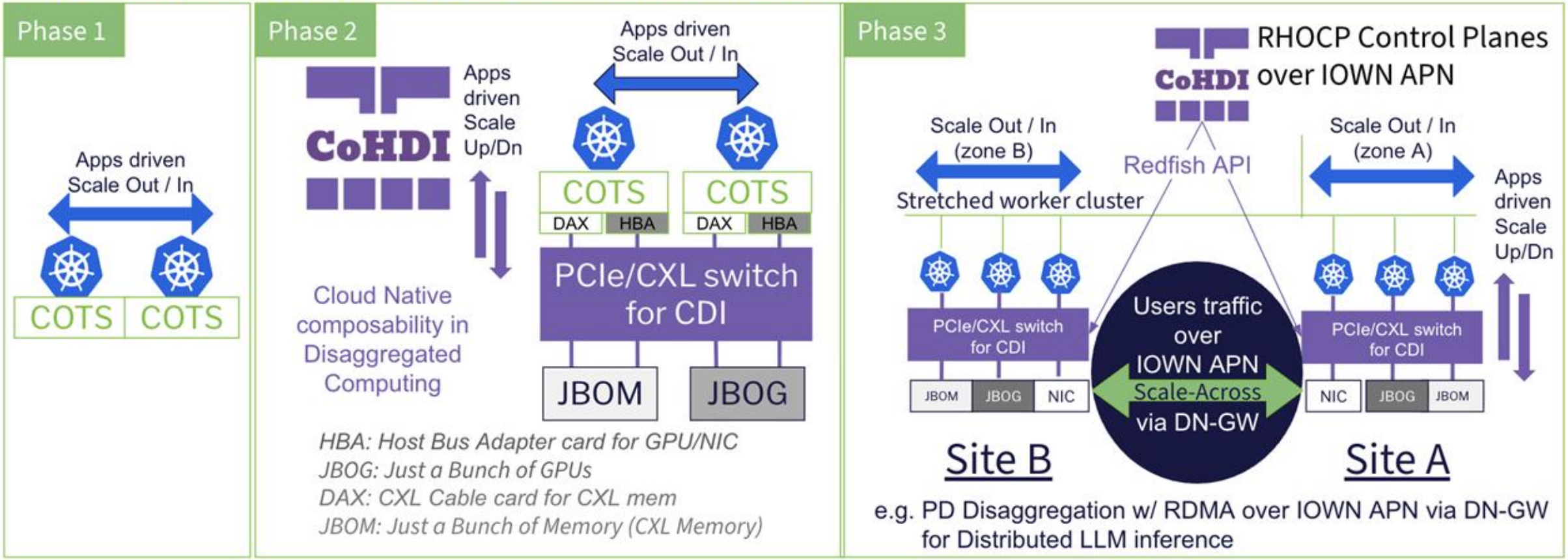
Implication: Open and interoperable optical interfaces are needed between storage-side optics and APN transport.

https://www.fiberlabs.co.jp/tech-explan/wavelength/ (2025-02-08 Referenced and Translated by NTT DATA)

AICC Compute-Across Models over the All-Photonics Network



Transitioning from COTS to Next-Generation Computing Systems for AICC with scale-across - IOWN infrastructure PoC



https://github.com/CoHDI

CoHDI catalog for Fujitsu(FSAS) Primergy-CDI

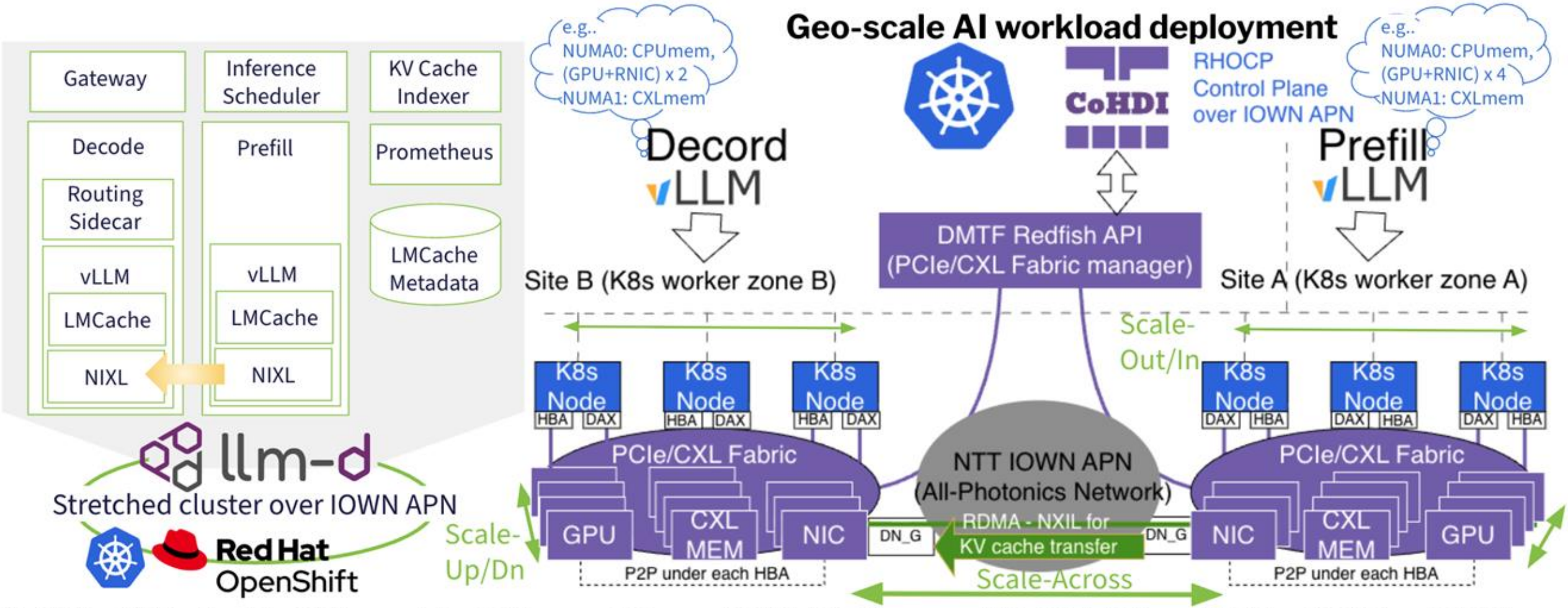
https://catalog.redhat.com/en/search?q=CoHDI

MHS (Modular Hardware System) is Under investigation

Ref https://www.cncf.io/projects/llm-d/ https://www.cncf.io/projects/cohdi/

Distributed LLM Inference on the IOWN APN/DN PoC

Prefill-Decode Disaggregation Connecting Regional and Telco Clouds via All-Photonics Network



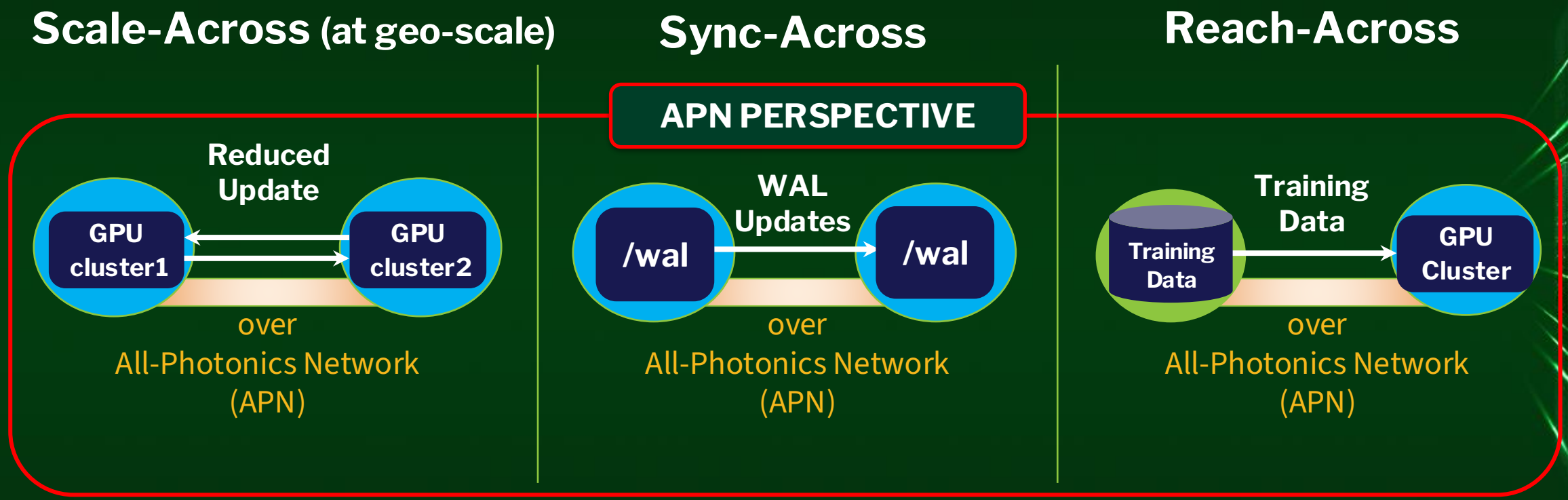
Multi-tier KV Cache: T1- GPUmem, **T2- CPUmem+CXLmem(DAX)**, T3- Storage NIC-GPU: P2P on HBA for GPU Direct RDMA

Optical SmartNIC for AI Infrastructure

APN perspective

AICC Compute-Across Models over the All-Photonics Network

Junichi Sugiyama, 1Finity



One cross-cutting layer across Scale-Across, Sync-Across and Reach-Across

Focus: **predictable** · **observable** · **interoperable** endpoint-to-optics behavior

Different Workloads, One Common Requirement

Toward OCP-Aligned Scale-Across AI Infrastructure | APN perspective

Lessons from preceding AICC talks: Scale-Up → Scale-Out → Scale-Across

Scale-Across

Cross-site KV / RDMA enters the inference path.

Fine-grained movement

Reach-Across

Pipelining can hide much of the network delay.

40 km: +1.9% | 100 km: 94%

Sync-Across

Distance remains visible in replication time.

Predictability enables design

Different workloads. Different distance sensitivity.

Common need: predictable, observable, interoperable endpoint-to-optics behavior.

Photonic Smart NIC: One Implementation Path

Reduce avoidable layering, buffering and RTT-dependent penalties at the long-reach endpoint

Endpoint-to-optics architecture

Workload GPU / Storage → **Programmable Photonic Endpoint** → Coherent Optics → APN

Simplified transport path

Reduce layered and RTT-dependent overhead.

Hardware acceleration

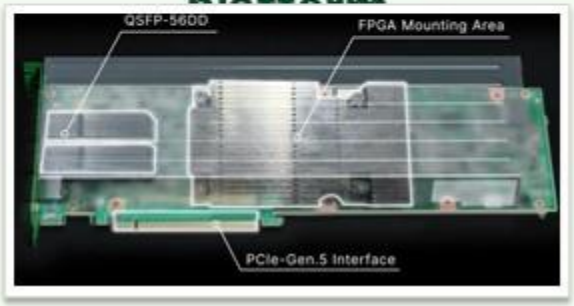
Move setup and data-path functions into hardware.

FPGA + telemetry

Expose telemetry and test pre-standard functions.

Not eliminating propagation delay—reducing avoidable penalties and exposing predictable behavior.

1Finity prototype platform



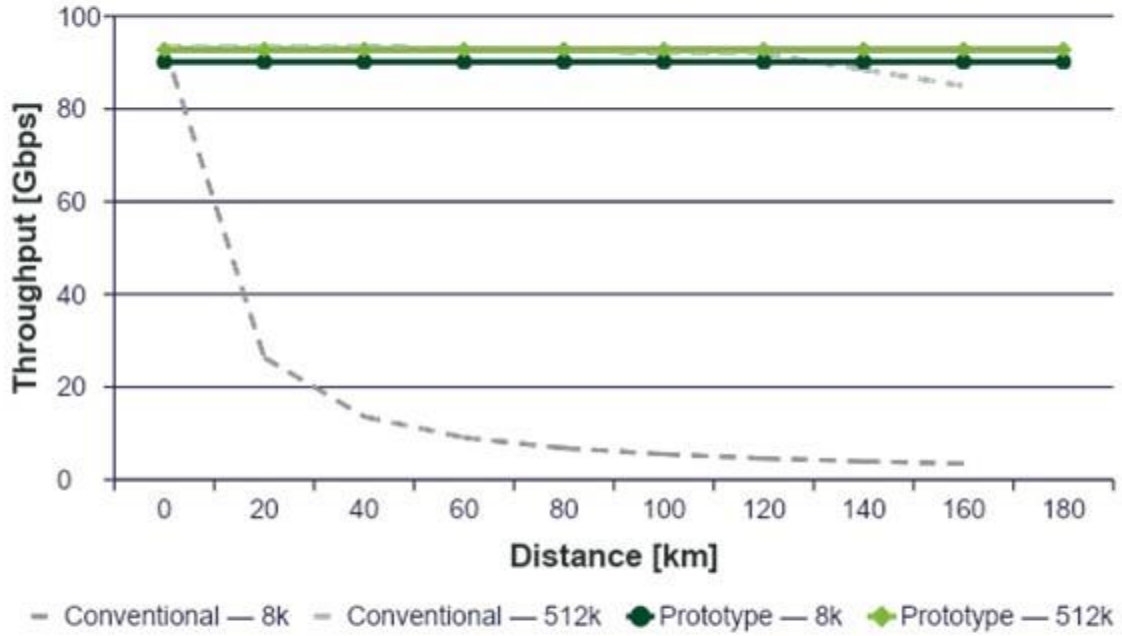


- FPGA-based endpoint
- Coherent pluggable support
- PoC / measurement platform

From Lab Evidence to Multi-Partner Validation

Measured behavior plus publicly announced field and ecosystem programs

Technical evidence



Stable throughput: ~90–93 Gbps through 180 km in the evaluated setup

Low variation: a separate latency-delta evaluation showed a tight distribution

~100G prototype; propagation delay remains; feasibility evidence, not a universal workload claim.

Public multi-partner programs

Regional distributed DC PoC

Kyushu Electric · IIJ · QTnet · Nautilus

APN + optical NIC + distributed DB

Launched Oct. 2025

Composable resource-sharing validation

Accton/Edgecore · Liqid

RDMA / NVMe-oF across multi-site optical paths

Public collaboration / SC25

Watt-Bit / Physical AI field PoC

Nautilus · IIJ · TEPCO PG · Chubu EPG · Fujitsu + partners

Remote GPUs + real-time 4K AI over 100 km+ optical NIC

Aug. 2026–Mar. 2027

Sources: Kyushu (Sep. 2025) | Edgecore (Nov. 2025) | Fujitsu consortium (Jul. 2026)

From device validation to cross-industry system validation.

From Partner Programs to Open OCP Contributions

Candidate OCP AICC path for a reusable Scale-Across Endpoint Profile

Public partner evidence → OCP AICC collaboration → Open, reusable outputs

1 Requirements / Endpoint Profile

- Workload classes and use cases
- Endpoint-to-optics interfaces
- Predictable behavior targets

2 Benchmark / Telemetry

- Distance, RTT and message size
- Throughput and tail latency
- Host–NIC–optics–APN visibility

3 Case Study / Tested Configuration

- Multi-vendor configurations
- Field evidence and lessons
- Repeatable validation methods

Complementary to OCP ESUN and UEC—focused on the long-reach endpoint-to-optics boundary.

1Finity can contribute: prototype platform · measured data · public use cases

Seeking OCP partners for an open, multi-vendor validation framework.

OCP path: Requirements · Benchmark · Case Study · Tested Configuration OCP Contribution Taxonomy (2026)

Cloud-Native Composability:

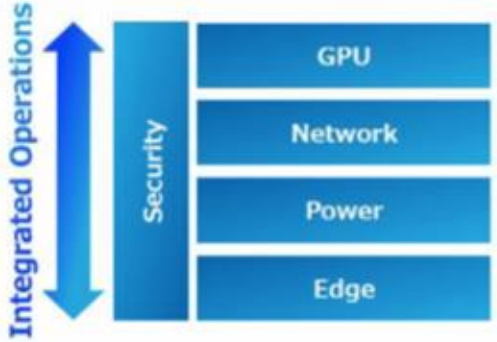
Future looking for AI-native Infrastructure by AICC

Enabling CoHDI(Cody) Across Regional and Telco Clouds via All-Photonics Network
CoHDI : Composable Hardware in Disaggregated Infrastructure

AI-native infrastructure that optimizes resources such as GPUs, networks, and power, while enabling operations extending to the edge

AIOWN

AI × IOWN



"AIOWN" is a trademark pending registration.



Optimization Domains

DC
Scalable and efficient data centers

Network
High-performance, low-latency connectivity

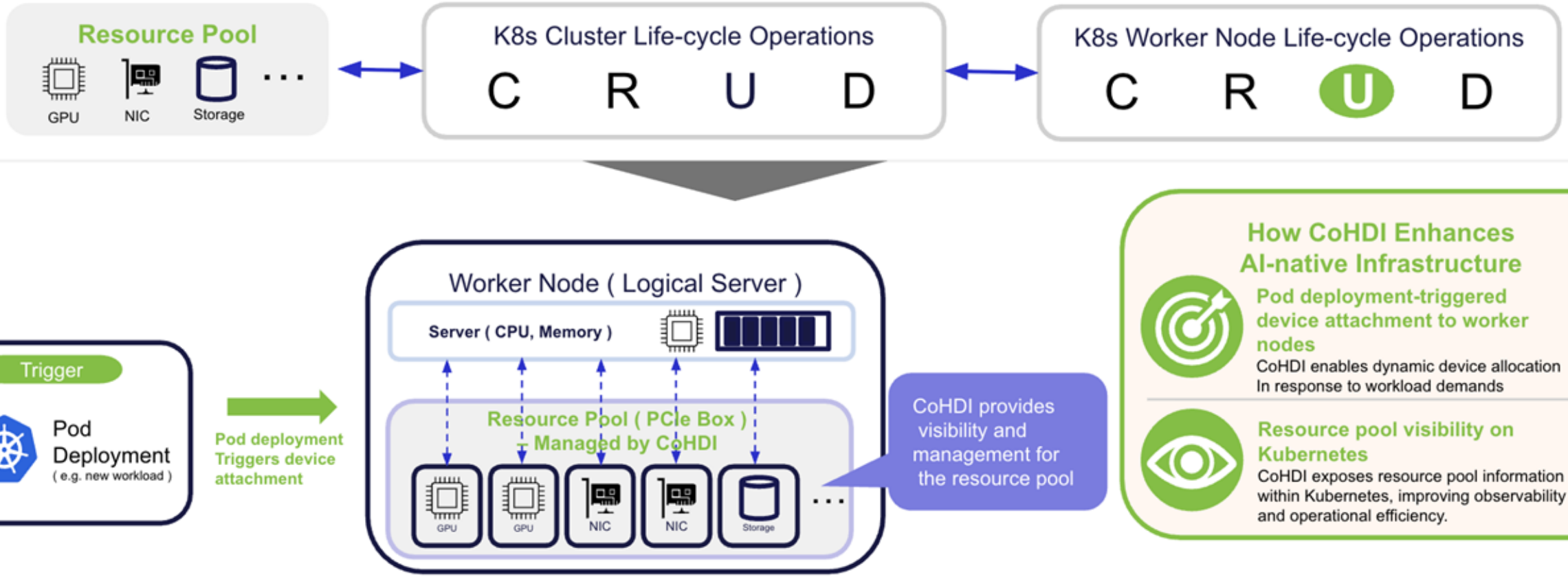
Resources
Optimal allocation of Compute resources

Energy Efficiency
Sustainable and efficient power usage

AI-native Infrastructure requires integrated optimization across Compute, Network and Energy domains

Use Case in AI-native Infrastructure w/ CoHDI

AI-native Infrastructure: Generalized requirements



By leveraging CoHDI, we bridge the gap between the resource pool and Kubernetes, enabling dynamic hardware utilization driven by Pod deployments.

Q & A

- *Reach-Across use case: Zero trust GPU sharing for data sovereignty*
- *Sync-Across use case: Financial service platform*
- *Scale-Across use case: Distributed LLM inference*
- *AICC use case with APN(All-Photonics Network) ready Smart-NIC*
- *Resource Pooling with Composable Hardware*

Thank You!

Delivering on the Next Wave of AI Inference

https://ocpstagingweb2.opencompute.org/community/aicc

MoU

